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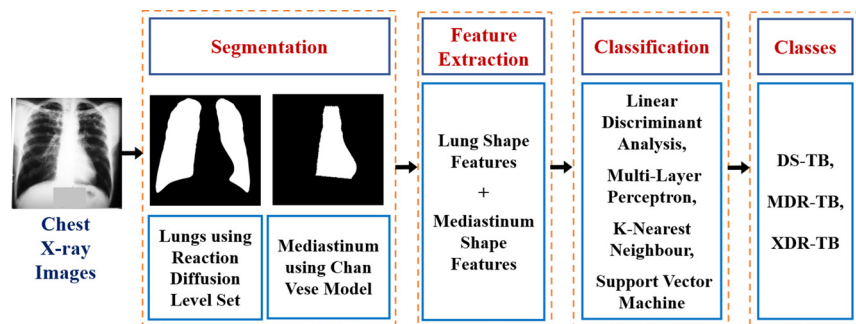
Evaluation of Diagnostic Value of Mediastinum for Differentiation of Drug Sensitive, Multi and Extensively Drug Resistant Tuberculosis Using Chest X-Rays

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HIGHLIGHTS

- Differentiation of Drug Sensitive and Resistant Tuberculosis using chest X-rays.
- Both lungs and mediastinum are precisely segmented using active contour models.
- Significant lung and mediastinum shape features are extracted for characterization.
- Multilayer perceptron and support vector machine classifiers perform better.
- Combination of mediastinum and lung features improves the diagnostic accuracy.

GRAPHICAL ABSTRACT



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ABSTRACT

Background and Objective: The rise of Drug Resistant Tuberculosis (DR TB), particularly Multi DR (MDR), and Extensively DR (XDR) has reduced the rate of control of the disease. Computer aided diagnosis using Chest X-rays (CXRs) can help in mass screening and timely diagnosis of DR TB, which is essential to administer proper treatment regimens. In CXRs, lungs and mediastinum are two significant regions which contain the information about the likelihood of DR TB. The objective of this work is to analyze the shape characteristics of lungs and mediastinum to improve the diagnostics accuracy for differentiation of Drug Sensitive (DS), MDR and XDR TB using computer aided diagnostics system.

Methods: The CXR images of DS and DR TB patients are obtained from a public database. The lung fields are segmented from the CXRs using Reaction Diffusion Level Set Evolution. Mediastinum is segmented from the delineated lung masks using Chan Vese model. The shape features from each lung and mediastinum masks are extracted and analysed. The discriminative power of individual and combination of both lung and mediastinum features are evaluated using machine learning techniques for classification of DS vs MDR, MDR vs XDR and DS vs XDR TB images. The performances of classifiers are compared using standard metrics.

Results: The proposed segmentation methods are able to delineate lungs and mediastinum from the CXR images. The extracted lung and mediastinum features are found to be statistically significant ($p < 0.05$) for differentiation of DS and DR TB conditions. Using the combination of both lung and mediastinum features, Multi-Layer Perceptron classifier achieves maximum F-measure of 82.4%, 81.0% and 87.0% for differentiation of DS vs MDR, MDR vs XDR and DS vs XDR, respectively.

Conclusion: Analysis of mediastinum along with the lungs in chest X-rays could improve the diagnostic performance for differentiation of drug sensitive and resistant TB conditions. The proposed methodology

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is able to differentiate DS, MDR and XDR TB, and found to be clinically relevant. Hence, this work is useful for computer-based early detection of DS and DR TB conditions.

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1. Introduction

Tuberculosis (TB) is a serious communicable disease that claims millions of lives each year worldwide. According to the 2020 World Health Organization (WHO) report, it is the foremost cause of death due to a single infectious agent [1]. The emergence of Drug Resistant (DR) TB, particularly Multi Drug Resistant (MDR), and Extensively Drug Resistant (XDR) has reduced the rate of control of the disease [2]. MDR TB is resistant to at least two of the efficient first line anti-TB drugs, namely isoniazid and rifampicin. XDR is MDR with additional resistance to a second line fluoroquinolone and an injectable drug [3]. Drug Sensitive (DS) TB is responsive to all anti-TB drugs that are recommended by WHO. In 2019, 10 million new TB cases were recorded worldwide. It is reported that 3.3% of new cases and 18% of previous cases are MDR TB, of which an estimated proportion of 6% are XDR TB [1]. Moreover, XDR and MDR TB has poorer treatment outcomes, higher treatment costs, and longer treatment time as compared to DS TB [2].

DR TB is commonly accompanied with delayed diagnosis, ineffective treatment or direct spread of drug resistant strains from one patient to another. Hence, to control the transmission of the disease there is an urgent need for the rapid detection of drug TB form and immediate treatment [3]. Currently, the drug susceptibility test is being carried out on the sputum sample for identification of the resistant status of the TB patient. However, the test requires a sophisticated laboratory and needs several weeks to obtain the results. The advance in real-time polymerase chain reaction has shortened the time for detection of DR TB. However, it produces non-conclusive results and is expensive to employ in resource-limited settings. Moreover, the tests need sputum sample, which can be cumbersome in mass screenings [2]. Therefore, early detection of DR TB is still a challenge. Automated diagnostic systems can significantly aid in the early detection and mass screening of the disease.

WHO highly recommends Chest X-rays (CXRs) as an essential imaging tool for early detection of TB conditions [4]. CXRs contain information that can be extracted by analysing the variations in significant anatomical structures to obtain the likelihood of DR TB infection [2]. Hence, it is a rapid tool in detection, surveillance and screening of DR TB. Further, digital CXRs are less expensive, easier to use and widely available [5]. Computer Aided Diagnosis (CAD) systems use image processing techniques to extract the relevant information from the CXRs [6]. These systems can help in automated detection and population screening of drug TB, especially in regions that lack radiological expertise. The main components of a CAD system are segmentation of Region of Interest (RoI), extraction of features from the RoI, and classification based on the features [7].

Lungs and mediastinum in CXRs are considered as two important anatomical regions for diagnosis of TB conditions. Mediastinum is the central region of thorax and is laterally surrounded by the lungs [8]. It has been reported that the common TB abnormalities such as mediastinal lymphadenopathy, pleural effusions, consolidation, cardiomegaly, fibrosis, infiltration, bronchial dilatation and mediastinal lesions lead to structural deformation of lungs and mediastinum [9–11]. In CXR images, pleural effusion appears as indistinctness in shape of medial and lateral pulmonary sulci, while mediastinal lymphadenopathy causes enlargement of lymph nodes in the close proximity of mediastinum. In cardiomegaly, the

width of inferior mediastinum increases due to enlarged heart and bronchial dilatation results in mediastinum widening due to air way enlargement. These structural alterations serve as image biomarkers for shape characterization of lungs and mediastinum to detect TB.

Similarly, abnormalities such as thick-walled cavities, lesions along bronchi, bronchiectasis, emphysema, atelectasis, calcification and effusion are frequent in DR TB as compared to DS [12,13]. Mediastinal lymph nodes, pneumoconiosis and exudative lesions are observed to occur frequently in DS TB [12,13]. Hence, shape variations in lungs and mediastinum are considered as significant biomarkers for discrimination of drug TB conditions. Therefore, analysis of variations in mediastinum along with lungs could improve the diagnostic accuracy for the detection of DR TB. However, there are limited CAD studies that utilizes shape characteristics of mediastinum for differentiation of DS and DR TB.

Different techniques such as rule-based, classification-based, deformable methods and hybrid approaches have been employed for segmentation of lungs from the CXRs [14]. Recently, level set based methods have been extensively used for effective segmentation of anatomical structures from medical images. Reaction Diffusion Level Set Evolution (RDLSE) is an edge-based segmentation method that is stable, free from re-initialization and has boundary antileakage property [15]. This technique has been proven to be successful in extracting complex structures from images by forming trajectory of irregular topological variations. Moreover, this achieves sub-pixel accuracy to segment smooth contours. In literature, it has been employed for segmentation of lungs from CXRs, corpus callosum and lateral ventricles from magnetic resonance images with better validation performances [7,16,17].

Chan Vese (CV) model is a well-known region-based method that has been applied for segmentation of medical images with two regions which have distinctively different mean pixel intensities. It uses global image information and handles the topological changes better for effective segmentation [18]. It has been widely used for the segmentation of lungs from radiographic images, and brain structures from magnetic resonance images with better performances [19,20].

TB CXR images exhibit irregular shape characteristics in lungs and mediastinum [9,11,21]. The shape features provide significant information about the variations of the anatomical structures for differentiation of TB conditions. These features exploit local and global representation of the anatomical regions [22]. Standard shape features such as size, orientation, eccentricity, extent, centroid and bounding box extracted from the segmented lung masks have been used to mimic the expert's comparative analysis process. It has been reported that they are sufficient to capture the image details with high performance [22–24]. However, these lung features have not been utilized for differentiation of DS and DR TB conditions. Moreover, the mediastinum shape features are yet to be explored for diagnosis of the disease.

The discriminative power of these features has been evaluated using different types of classifiers for automated differentiation of TB conditions. A good classifier produces a better decision boundary to discriminate the image classes. Some of the proposed CAD systems have employed K-Nearest Neighbour (KNN), Multi-Layer Perceptron (MLP), Support Vector Machine (SVM) and Linear Discriminant Analysis (LDA) for differentiation of healthy, DS, and DR TB CXR images [2,5,14,25]. Therefore, as noted earlier, it is desir-

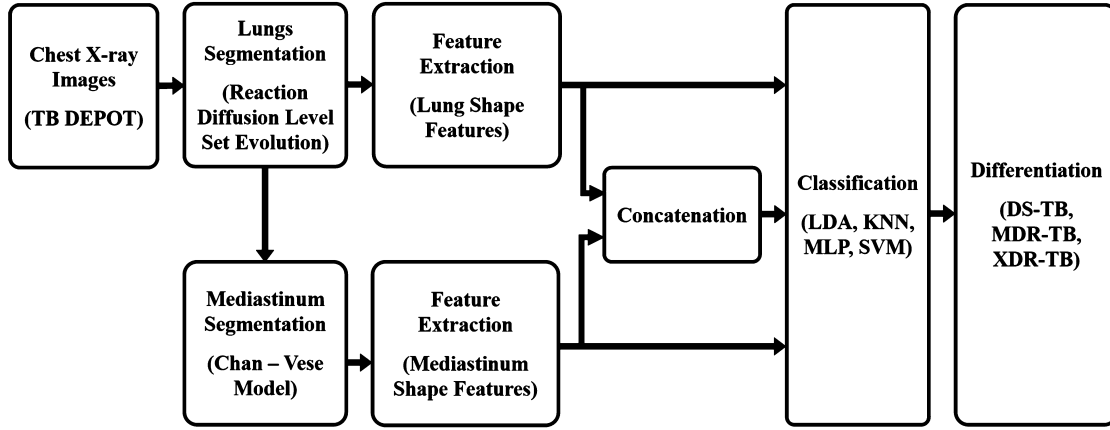


Fig. 1. Pipeline of the proposed methodology.

able to understand the variations in mediastinum along with lungs, using a CAD system.

The objective of this work is to analyze the shape characteristics of lungs along with mediastinum to improve the diagnostics accuracy for differentiation of DS, MDR and XDR TB using a CAD system. The novelty of this work is to employ Chan Vese model for segmentation of mediastinum from both DS and DR TB CXR images. RDLSE method is adopted to segment the lungs from these images. Further, the shape features are extracted from both the regions, and are fused. The discriminative power of these features is assessed using machine learning approaches. The rest of the paper is organized as materials and methods, results and discussion, and conclusions followed by references.

2. Materials and methods

The pipeline of the proposed method is illustrated in Fig. 1. The chest X-ray images of DS, MDR and XDR TB patients are obtained from a public database [26]. The lung fields are segmented from the CXRs using RDLSE method, followed by segmentation of mediastinum using CV model. Shape features from both lung and mediastinum masks are extracted and analysed to differentiate the TB conditions. The discriminative power of individual and combination of both lung and mediastinum features are assessed utilizing machine learning techniques such as LDA, KNN, MLP and SVM. The classifiers are employed for differentiation of DS vs MDR, MDR vs XDR and DS vs XDR TB images, and the performances are compared using standard metrics such as accuracy, sensitivity, precision, F-measure and Area Under the receiver operating characteristic Curve (AUC).

2.1. TB CXR image database

300 TB CXR images (100 DS, 100 MDR and 100 XDR TB) are obtained from Tuberculosis Data Exploration Portal (TB DEPOT), a publicly available database [26]. TB DEPOT is developed by the National Institute of Allergy and Infectious Diseases (NIAID) to address the complex etiology of TB conditions. It leverages the data collected from NIAID's TB Portals Program (TBPP). The TBPP database is a global collaboration that curates a large collection of physicians validated, deidentified and accessible TB patient data [27]. TB DEPOT displays these patient data and facilitates studies across an extensive range of clinical, genomic and radiological factors. Permission to access the database was obtained by signing a memorandum of understanding. The CXR images are obtained in grayscale. Then, these are resized to a fixed pixel resolution of 512×512 by using bicubic interpolation. This technique improves the

signal to noise ratio by suppressing noisy characteristics to execute precise segmentation [14].

2.2. Segmentation of lung fields using reaction diffusion level set evolution (RDLSE)

The lung fields from the CXRs are segmented using RDLSE method. RDLSE is an edge-based active contour model, where a basic curve can be represented as Eq. (1) [15].

$$C(x, t) : [0, 1] \times [0, \infty) \rightarrow \mathbb{R}^2, \quad x \in [0, 1] \text{ and } t \in [0, \infty] \quad (1)$$

where x is the spatial parameter and t is the time generated by movement of initial contour.

The curve under evolution is given by Eq. (2).

$$\frac{\partial C(x, t)}{\partial t} = F \vec{N} \quad (2)$$

where F is the force function to control movement of contour and $\vec{N}$ is the normal vector to C .

The basic rate equation can be represented as in Eq. (3).

$$\frac{\partial \phi}{\partial t} = \text{Reg}(\phi) + F\delta(\phi) \quad (3)$$

where $\text{Reg}(\phi) = \alpha \text{div}[r(\phi)\nabla\phi]$, α is a constant, $r(\phi)$ is equation of conventional diffusion rate and $F\delta(\phi)$ is the reaction term.

The conventional diffusion rate equation suffers from boundary leakage issue. Hence, to avoid this issue, a diffusion term, $\varepsilon\Delta\phi$ is added in the level set to obtain a stable equation known as RDLSE, which is provided in Eq. (4).

$$\frac{\partial \phi}{\partial t} = \varepsilon\Delta\phi - \frac{1}{\varepsilon}L(\phi), \quad x \in \Omega \subset \mathbb{R}^n \quad (4)$$

where ε represents a constant value, Δ is Laplacian operator, and $-\frac{1}{\varepsilon}L(\phi) = F\delta(\phi)$ is the reaction term. F can be expressed as given in Eq. (5).

$$F = \text{div} \left(\frac{g(|\nabla I_\sigma|) \nabla \phi}{|\nabla I_\sigma|} \right) + v g(|\nabla I_\sigma|) \quad (5)$$

where $g(|\nabla I_\sigma|)$ is an edge indicator, I_σ is the smoothed image, v is a constant and g can be mathematically expressed as in Eq. (6).

$$g = \frac{1}{1 + |\Delta I_\sigma|} \quad (6)$$

In the current work, α is set to 0.7 and the number of iterations is fixed at 800 for effective lung segmentation [16].

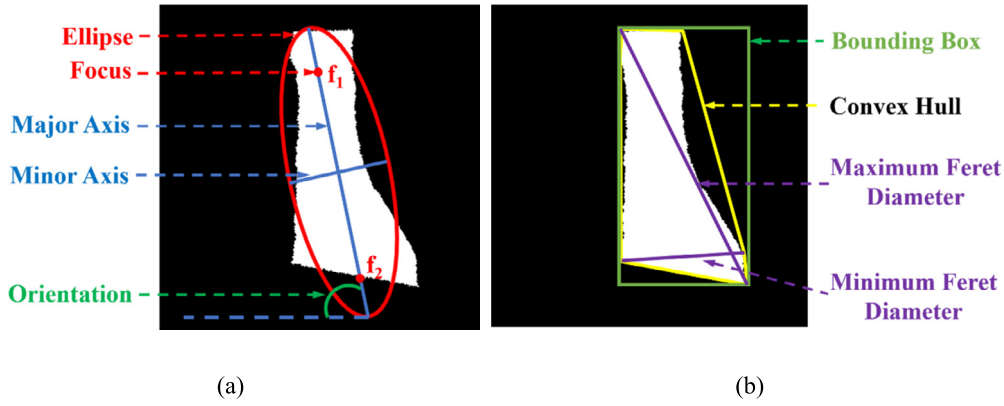


Fig. 2. Illustration of shape-based features (a) Ellipse fit and (b) Convex hull and bounding box of a representative binary image.

2.3. Segmentation of mediastinum using Chan Vese model (CV model)

The mediastinum is segmented from the delineated lung masks using CV model. It is a region-based active contour model. In this method, for an input image $I(x)$ in the domain Ω and a closed curve C , the energy function is defined as given in Eq. (7) [18].

$$E^{CV}(c_1, c_2, C) = \lambda_1 \int_{in(C)} |I(x) - c_1|^2 dx + \lambda_2 \int_{out(C)} |I(x) - c_2|^2 dx \quad (7)$$

where c_1 and c_2 are constants that represent the average intensities inside and outside of C , respectively, $\lambda_1, \lambda_2 > 0$ are the coefficients which controls the drive force inside and outside C , respectively.

In order to control the smoothness of boundary, the regularizing terms length and inner area of C is added. The modified energy function is defined as in Eq. (8).

$$E^{CV}(c_1, c_2, C) = \lambda_1 \int_{in(C)} |I(x) - c_1|^2 dx + \lambda_2 \int_{out(C)} |I(x) - c_2|^2 dx + \mu \text{Length}(C) + \nu \text{Area}(in(C)) \quad (8)$$

where $\mu \geq 0$ is a scaling parameter that controls the evolution and $\nu \geq 0$ is the speed of propagation.

Representing the contour C as a zero-level set, $C = \{x \in \Omega | \phi(x) = 0\}$, the energy function can be given as shown in Eq. (9).

$$E^{CV}(c_1, c_2, \phi) = \lambda_1 \int_{\Omega} |I(x) - c_1|^2 H(\phi(x)) dx + \lambda_2 \int_{\Omega} |I(x) - c_2|^2 (1 - H(\phi(x))) dx + \mu \int_{\Omega} \delta(\phi(x)) |\nabla \phi(x)| dx + \nu \int_{\Omega} H(\phi(x)) dx \quad (9)$$

where $\delta(\phi)$ is Dirac and $H(\phi)$ is Heaviside function.

Minimizing the above energy function using steepest descent method yields the corresponding variational level set equation given as Eq. (10) [19].

$$\frac{\partial \phi(x, t)}{\partial t} = \delta(\phi) \left[-\lambda_1 (I(x) - c_1)^2 + \lambda_2 (I(x) - c_2)^2 + \mu \text{div} \left(\frac{\nabla \phi}{|\nabla \phi|} \right) - \nu \right] \quad (10)$$

where ∇ represents the gradient operator. Here, in the CV model the number of iterations is empirically fixed at 900 for precise segmentation of the mediastinum.

2.4. Feature extraction

To characterize the shape variations in lungs and mediastinum due to DS, MDR and XDR TB conditions, 12 shape-based features each from the segmented lung and mediastinum masks are extracted. The graphical representation of the features is illustrated in Fig. 2. A binary image with an ellipse that has the same normalized second central moments as the image is shown in Fig. 2 (a). The convex hull and the bounding box of the given image is shown in Fig. 2 (b). Convex hull of a region specifies the smallest convex polygon that contains the whole region. Bounding box is the smallest rectangular border that can fully enclose the given image as shown in Fig. 2 (b) [22–24]. The description of extracted features is provided below.

Area: In a binary image, the area is defined as the sum of pixels in the considered region. The pixels within the region are marked as $f(x, y) = 1$, and the pixels outside the region as $f(x, y) = 0$. The area can be calculated as:

$$\text{Area} = \sum_{(x,y) \in R} f(x, y) \quad (11)$$

The area feature measures the size of lung and mediastinum masks.

Major and Minor axis length: Return scalar values in pixels that specify the length of major and minor axis of ellipse as shown in Fig. 2 (a). In the figure, larger value of major axis length signifies that the shape is taller and larger minor axis length shows that it is wider.

Eccentricity: It is defined as the ratio of distance between the two foci (f_1, f_2) and major axis length of the ellipse. From Fig. 2 (a), the eccentricity can be calculated as shown in Eq. (12)

$$\text{Eccentricity} = \frac{|f_1 f_2|}{\text{Major axis length}} \quad (12)$$

Eccentricity measures the extent of roundness of lungs or mediastinum and tends to zero for a rounded field and tends to one for a squashed shape.

Orientation: Returns a scalar value representing the angle between x -axis and major axis of ellipse as presented in Fig. 2 (a). It shows the inclination of the structure, where 0° and 90° represents the horizontal and vertical ellipses, respectively.

Convex area: Represents the total number of pixels present inside the convex hull of the considered region as shown in Fig. 2 (b).

Equivalent diameter: Defined as the length of diameter of a circle that has the same area as the considered image. It is calculated as given in Eq. (13).

$$\text{Equivalent diameter} = \sqrt{\frac{4 \times \text{Area}}{\pi}} \quad (13)$$

Solidity: Proportion of the number of pixels in the considered region to the number of pixels present in the convex hull of the region. It is computed as given in Eq. (14).

$$\text{Solidity} = \frac{\text{Area}}{\text{Convex area}} \quad (14)$$

Solidity of 1 signifies that the considered region is a solid object with regular boundary, while value less than 1 signifies that the region has an irregular boundary.

Extent: A scalar value that specifies the ratio of number of pixels in the region of interest to the number of pixels in its bounding box. It is computed as shown in Eq. (15).

$$\text{Extent} = \frac{\text{Area}}{\text{Bounding box area}} \quad (15)$$

Extent represents the rectangularity of the region. Extent of 1 signifies that the region is a rectangle.

Perimeter: Defined as the length of the outer boundary of the region. It is computed by calculating the distance between each adjoining pair of pixels around the border.

Maximum and Minimum feret diameter: Defined as the maximum and minimum distance between two boundary points present on the antipodal vertices of the convex hull that can enclose the object. These features are illustrated in Fig. 2 (b). Larger value of maximum feret diameter signifies that the shape is taller, while larger minimum feret diameter indicates that the shape is wider.

These features are extracted from both the lung and mediastinum masks and are statistically analyzed for differentiation of DS and DR TB conditions. The normality of the extracted features is analyzed using Shapiro - Wilk test. For non-normally distributed features, the statistical significance ($p < 0.05$) analysis is performed using Wilcoxon rank sum test for differentiation of DS vs MDR, MDR vs XDR and DS vs XDR TB.

2.5. Classification of TB conditions

In the current work, four different classifiers are employed for differentiation of DS vs MDR, MDR vs XDR and DS vs XDR TB CXRs by using individual and combination of both lung and mediastinum features. Two widely used parametric classifiers, namely LDA and MLP, and two non-parametric classifiers, namely KNN and SVM are employed. Five-fold cross validation is performed and the average scores of the folds are reported as the final performance values.

Linear Discriminant Analysis (LDA)

LDA is a supervised classifier that finds an optimal linear transformation of the input features to maximize the separation between two classes. The best transformation exhibits a maximum distance between the elements of classes while a minimum distance within a class is maintained. It creates a linear combination of features that can be used as a linear classifier. It finds a projection direction that maximizes the ratio of variance between classes to the variance within the class to ensure maximal separability [6]. In this study, the discriminant type is selected as linear and the value of regularization parameter, gamma is set at 0 to use the unrestricted, empirical covariance matrix. The algorithm applies minimal regularization for inversion of a singular covariance matrix.

Multi-Layer Perceptron (MLP)

MLP is a feed forward neural network-based classifier that is trained with back propagation algorithm. It divides the neurons into three main layers known as input, hidden and output. The features are applied to the input layer and transferred to the hidden layer where each neuron is activated using an activation function. The result of the operation is obtained in the output layer. Here, the non-linear sigmoid function is used as the activation function. Learning rate of 0.3 and momentum of 0.2 is considered. The number of neurons in the input layer and output layer is set as the number of input features and output classes, respectively [14,28]. The number of neurons in the hidden layer is empirically varied from 10 to 50, and set to 30 based on the obtained maximum accuracy.

K-Nearest Neighbour (KNN)

KNN is a supervised classifier used to differentiate the feature sets. This uses the feature similarity value for predicting the class of test data points. In this, a class is allocated to the test data point depending on their closeness with the points of training set. It identifies K data points from the training set which have minimum distance from the test data point. Based on the most common class of the identified K data points, the algorithm allocates a class [5,7]. Euclidean distance is used as the distance metric with equal weight. The number of nearest neighbours, K is analytically varied from 1 to 9, and fixed to 5 based on classification accuracy.

Support Vector Machine (SVM)

SVM is a supervised classifier that creates a hyperplane to classify two classes. It finds the best separating hyperplane with largest distance between the feature vectors of the classes in the training data. The test feature is assigned with a class by calculating the distance of test data point from the computed hyperplane. This can handle high dimensional feature sets with fast learning capacity. This classifier is also reported to be robust and achieves good performance even with a small training set [2,7]. The linear kernel function is chosen and a subsampling based heuristic procedure is employed for selection of kernel scale value. The value of box constraint parameter that represents the soft-margin penalty is chosen as 1, as higher value can decrease the number of support vectors and increase the training time.

2.6. Performance evaluation

Performance of the employed classifiers is analyzed using different evaluation metrics such as accuracy, sensitivity, precision, F-measure and AUC [29,30]. The accuracy shows the proximity of the result to the actual classes by the relationship between the set of possibilities of errors and correctness, and is presented by Eq. (16).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \quad (16)$$

where TP (True Positive) represents the total number of positive class CXRs correctly identified, FP (False Positive) is the total count of negative class images falsely recognized as positive class, TN (True Negative) is the total number of negative class CXRs correctly classified and FN (False Negative) represents the total count of positive class images falsely identified as negative class.

Sensitivity represents the proportion of the number of CXRs classified as positive class among the actual number of positive CXRs and is presented by Eq. (17).

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (17)$$

The precision is the probability of the true positive CXRs in relation to all the CXRs classified as positive and is presented by Eq. (18).

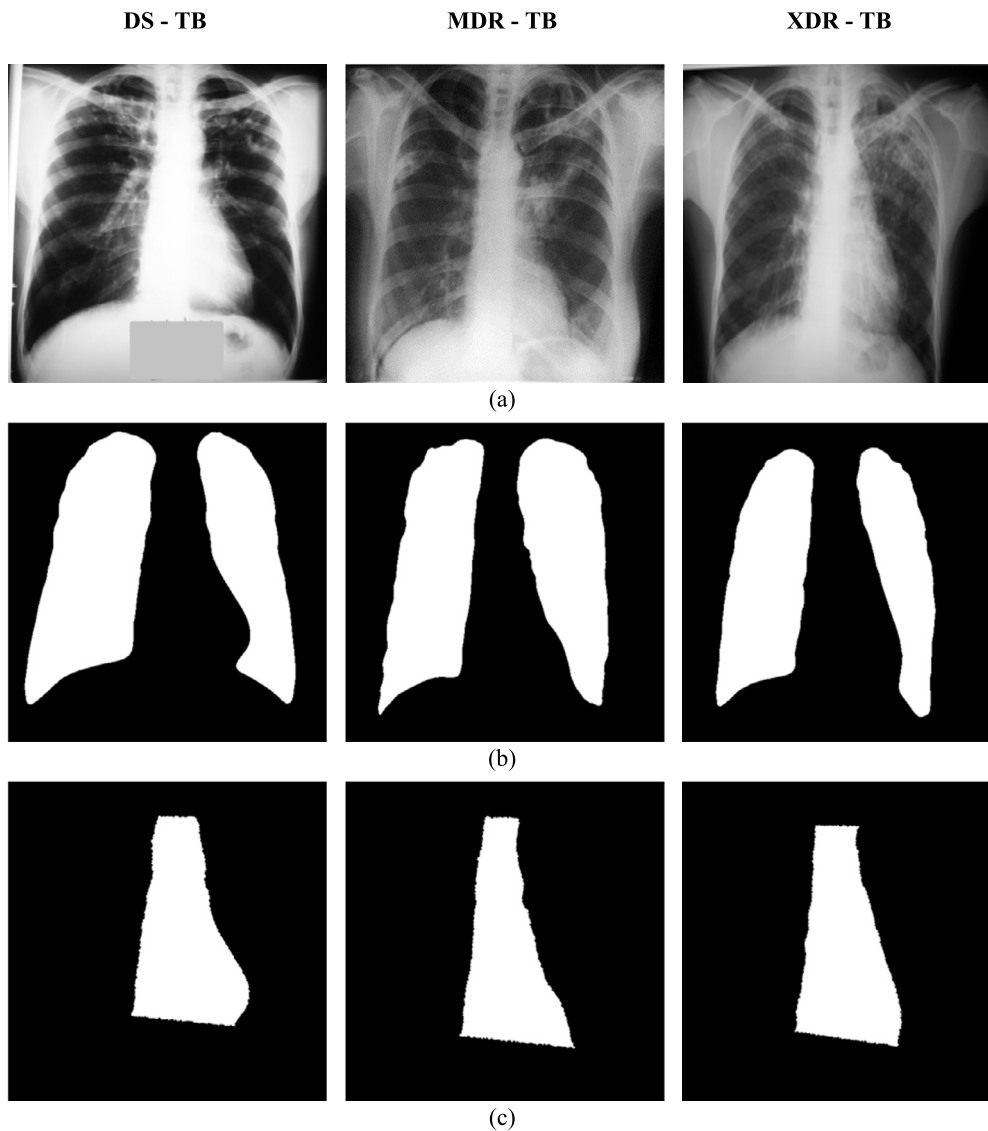


Fig. 3. Representative images of (a) Raw CXRs, (b) Segmented lung masks and (c) Segmented mediastinum masks.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (18)$$

Considering the risk of imbalance of precision and sensitivity, F-measure calculates the weighted mean harmonic value between precision and sensitivity and is presented by Eq. (19).

$$\text{F-measure} = \frac{2 \times \text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}} \quad (19)$$

Receiver Operating Characteristic (ROC) analysis is widely used for analysis and comparison of diagnostic accuracy of classifiers. ROC curve is a graphical representation of classifier performance generated by plotting the true positive rate in y-axis against the false positive rate in x-axis, at various threshold settings [30]. This can provide a better measure of performance than the scalar measures by depicting the trade-off between hit rates and false alarm rates. AUC is a single scalar value that is calculated from the ROC analysis and is used to make effective comparisons between the classifiers. For mass screening of the disease, it is desirable to minimize the false negative cases even at the cost of false positives [22]. The analysis is performed using a system with Intel i5 processor, 8 GB RAM and MATLAB® 2021a software.

3. Results and discussion

The representative raw chest X-ray images of DS, MDR and XDR TB patients are shown in Fig. 3 (a). Visually, it is observed that both lungs and mediastinum exhibit structural dissimilarity for different TB conditions. For DS TB, the area of lung fields appears to be larger than MDR and XDR images. The mediastinum also seems to be larger in DS TB compared to MDR and XDR. The lungs area appears to be the smallest for XDR TB while mediastinum area seems to be similar for MDR and XDR. However, the lungs and mediastinum boundaries are indefinite and unclear. Hence, detection of lungs and mediastinum shape through visual examination is challenging. In addition, the ill-defined boundaries and structural differences impose challenges for accurate segmentation of these regions.

Fig. 3 (b) presents the representative lung masks for DS, MDR and XDR TB images, segmented using RDLSE method. It is observed that RDLSE is able to capture the significant edges of lungs that could aid in precise segmentation of the region. It is also found that the visual observation regarding the shape of lung masks correlates with the observations from raw CXRs. The representative mediastinum masks for DS, MDR and XDR TB chest X-rays segmented using CV model are shown in Fig. 3 (c). It is found that

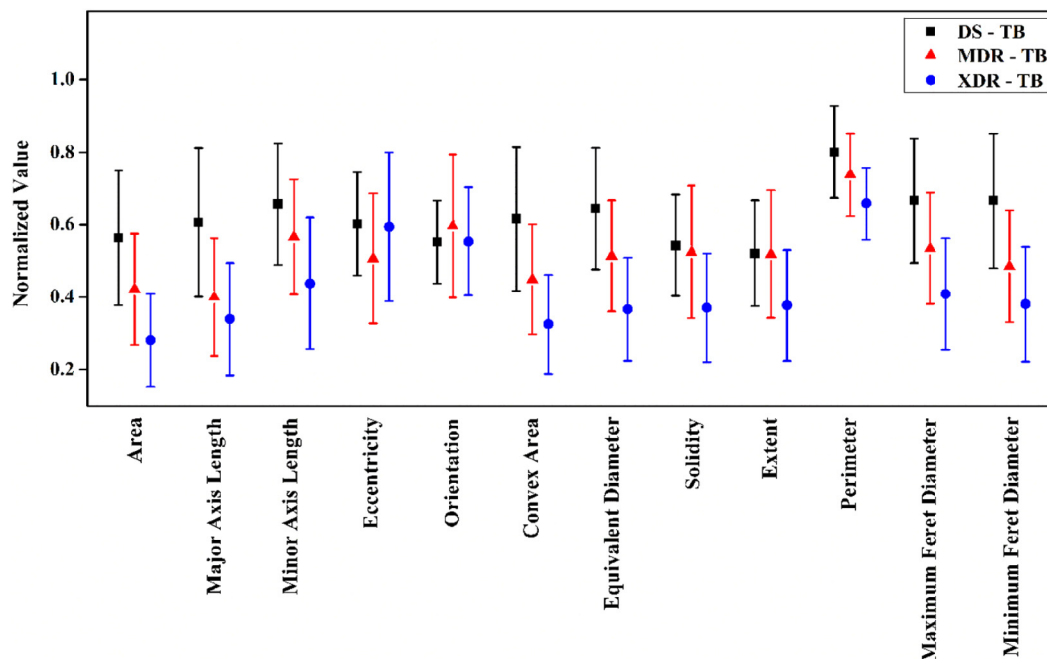


Fig. 4. Variation of lung features between DS, MDR and XDR TB CXRs.

the employed model not only precisely segments the mediastinum but also retains the shape variations of the region. From the masks, difference in shape and size of the mediastinum is observed between DS, MDR and XDR TB conditions. However, visual inspection of these shape characteristics is difficult. Hence, the use of shape features to quantify these structural dissimilarities can aid in the differentiation of TB conditions.

The variation in lung shape features for differentiation of DS, MDR and XDR TB is presented in Fig. 4. It is observed that the features such as area, major axis length, minor axis length, convex area, equivalent diameter, solidity, extent, perimeter, maximum feret diameter, minimum feret diameter have significantly greater mean value for DS TB than DR TB.

It can be observed that for DS TB the lungs are having larger size as compared to DR TB conditions. This indicates that with increasing severity of TB condition, the lungs undergo structural deterioration. This might be due to frequent occurrence of multiple cavities, nodules, whole-lung involvement, collapsed lung, bronchiectasis, emphysema, calcification, pleural effusion and lung volume loss in DR TB than in DS TB [11,12]. In XDR TB, the mean lungs size is found to be even smaller than MDR TB. This can be attributed to frequent occurrence of nodules, reticulo-nodular densities, consolidation and cavities in XDR patients compared to MDR TB [13,31].

The ability of the extracted mediastinum shape features to differentiate DS, MDR and XDR TB is graphically analysed in Fig. 5. It is found that for the features namely, area, major axis length, minor axis length, convex area, equivalent diameter, perimeter, maximum feret diameter and minimum feret diameter, the mean values are significantly more in DS TB than DR TB conditions. This indicates that for the DS TB condition the mediastinum appears to be larger as compared to DR TB. This can be attributed to the presence of frequent calcified lymph nodes, large nodules and ground-glass opacity lesions along the mediastinum in the DS group than in the DR [11,12]. The presence of these abnormalities in DS group can be associated with enlargement of mediastinum.

For MDR and XDR TB, the area, orientation and equivalent diameter are found to have similar mean values, which indicate that there is no significant change in the area of mediastinum due to

these conditions. However, it is observed that major axis length, eccentricity, convex area, perimeter and maximum feret diameter are significantly larger for MDR TB than XDR TB. This shows that the length of mediastinum in MDR TB is greater than XDR, which could be due to presence of frequent consolidation, thick-walled multiple cavities and bronchiectasis in MDR TB [13,31]. However, it is also found that the mean of minor axis length and minimum feret diameter are larger for XDR TB as compared to MDR TB, which are associated with widening of the mediastinum. This can be attributed to frequent presence of tree-in-bud appearance and lobular consolidations along the mediastinum in XDR TB [31]. The significant differences in the mean values suggest that these shape features can aid in the effective discrimination of DS, MDR and XDR TB conditions.

The distribution of lungs and mediastinum area feature of DS and XDR TB images along with their mean and Standard Deviation (SD) are graphically shown in Fig. 6 (a-b), respectively. The figure shows that, the area of both lungs and mediastinum of XDR TB images has significantly low magnitude compared to DS TB images. It can also be observed that, the area features of DS TB images are widely scattered than the XDR TB images. From Fig. 6 (a), less overlap is seen in the lung area feature and both the classes are found to be better differentiated. Whereas, from Fig. 6 (b) a moderate amount of overlap is observed in the mediastinum area feature space. However, a significant distinction is observed between the mean values. Hence, the area feature is able to differentiate the shape variations in lungs and mediastinum that occur due to these TB conditions.

The statistical analysis of the 12 shape features extracted from the lung and mediastinum masks for three sets of binary classifications of TB conditions are presented in Table 1. For differentiation of DS vs MDR TB, the lung feature, namely major axis length is found to have maximum mean difference of 54.6% and the mediastinal feature, namely minor axis length has provided maximum difference of 50.9%. 10 features each from the lung and mediastinum are found to be statistically significant ($p < 0.05$). For differentiation of MDR vs XDR TB, the area of lung and major axis length of mediastinum show maximum variation of 33.5% and 50.3%, respectively. Eleven lung and nine mediastinum features are

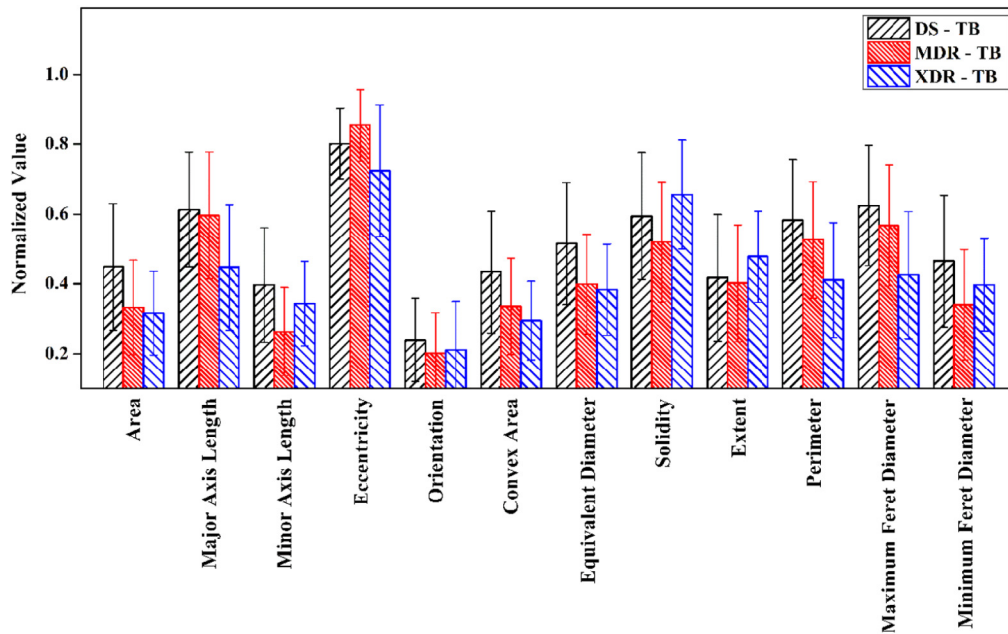


Fig. 5. Variation of mediastinum features between DS, MDR and XDR TB CXRs.

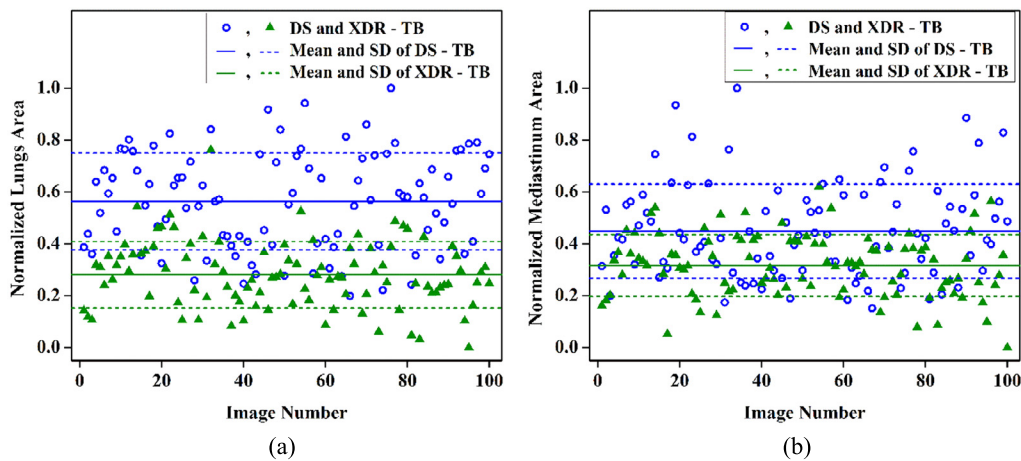


Fig. 6. Scatter plot representation of (a) Lung area and (b) Mediastinum area feature for DS and XDR TB.

found to be statistically significant ($p < 0.05$) for differentiation of MDR vs XDR TB condition.

For differentiation of DS vs XDR TB images, the mean area of lungs for DS TB images is observed to be double than the area of XDR TB CXR images. It is also found that the convex area of mediastinum has provided maximum mean difference of 49.7%. 10 lung and 11 mediastinum shape features are found to be statistically significant ($p < 0.05$) for classification of DS and XDR TB conditions. It is found that at least nine features each from lung and mediastinum are found to be statistically significant for all the three binary classifications of TB conditions. Hence, it can be inferred that the extracted shape features from both the lung and mediastinum illustrate the ability of capturing the shape variations for efficient differentiation of DS, MDR and XDR TB conditions.

Further, the classification ability of the extracted lung and mediastinum features is evaluated using LDA, KNN, MLP and SVM classifiers. Fig. 7 depicts the accuracy and ROC analyses of classifiers using individual, and combination of both lung and mediastinum features for differentiation of DS vs MDR, MDR vs XDR and DS vs XDR TB. From the figure, it is observed that the com-

bined lung and mediastinum feature set performs better than the individual features for all the three binary classifications.

Examining Fig. 7 (a), it is found that the combined feature set has achieved maximum accuracy of 82.5% using MLP classifier for differentiation of DS vs MDR TB. Individually, accuracy of 77.5% is achieved using lung features and MLP, while mediastinum features have provided maximum accuracy of 73.5% using SVM. Further, the classifiers with maximum accuracy are considered for ROC analysis. From Fig. 7 (b), it is found that the combined features have achieved maximum AUC of 87.3% for differentiation of DS vs MDR TB using MLP. Higher AUC corresponds to attainment of better degree of separability by the classifier for differentiation of the classes. Better performance of MLP and SVM classifier could be due to good feature learning and generation of optimum separating surface for differentiation of the classes. Moreover, in MLP, the network weights are tuned such that the sum-of-square error between the output and the actual value is minimized.

Fig. 7 (c) presents the plot of accuracy for differentiation of MDR and XDR TB images. It is found that the combination of lung and mediastinum features has achieved maximum accuracy of 81%

Table 1
Statistical analysis of extracted lung and mediastinum features.

Features	Mean difference (in %)					
	DS vs MDR		MDR vs XDR		DS vs XDR	
	Lungs	Mediastinum	Lungs	Mediastinum	Lungs	Mediastinum
Area	42.0*	38.3*	50.3*	5.1	100.0*	41.9*
Major Axis Length	54.6*	3.9	17.9*	33.5*	78.8*	37.2*
Minor Axis Length	20.9*	50.9*	29.2*	30.6*	50*	19.5*
Eccentricity	19.0*	11.8*	17.4*	18.1*	1.5	10.7*
Orientation	8.1*	19*	7.8	5.0	0.3	12.9
Convex Area	46.5*	29.5*	38.2*	13.7*	89.8*	49.7*
Equivalent Diameter	34.3*	32.4*	40.1*	4.2	76*	34.6*
Solidity	5.0	14.3*	41.7*	27.3*	46.6*	10.3*
Extent	0.6	3.9	37.4*	19.5*	38.1*	14.3*
Perimeter	8.5*	11.6*	12*	28.0*	22.4*	41.8*
Maximum Feret Diameter	30.9*	13*	30.7*	33.2*	63*	46.6*
Minimum Feret Diameter	53.3*	36.7*	27.5*	16.8*	75.2*	20*

* Indicates statistically significant features ($p < 0.05$).

Bold indicates the maximum mean difference among the features.

using MLP. Individually, accuracy of 78.0% and 80.5% is obtained using lung and mediastinum features, respectively. It is also found that both mediastinum and combined feature set have yielded AUC of greater than 86.6% as shown in Fig. 7 (d). Hence, mediastinum features are found to be more effective in differentiation of MDR vs XDR TB. The accuracy for differentiation of DS and XDR TB is shown in Fig. 7 (e). Using MLP, maximum accuracy of 87.0% is achieved using the combined features, while accuracy of 85.5% and 75.5% are found using individual lung and mediastinum features, respectively. Maximum AUC of 93.5% is obtained using the combined features as shown in Fig. 7 (f). Hence, it can be inferred that consideration of mediastinum features along with lung features aids in improvement of the diagnostic accuracy for differentiation of drug sensitive and resistant TB conditions.

The maximum performance measures obtained for each binary classification using individual and combination of both lung and mediastinum features are tabulated in Table 2. For differentiation of DS vs MDR TB, the combined feature set has achieved maximum accuracy, sensitivity, precision and F-measure of 82.5%, 82.5%, 83.1% and 82.4%, respectively, using MLP classifier. Better performance measures show good proportion of correct predictions and a smaller number of false predictions for both the classes. High F-measure score indicates, good balance of sensitivity and precision of the classifiers. For differentiation of MDR vs XDR and DS vs XDR TB, the combined feature set have obtained maximum classification measures of greater than 81% and 87%, respectively, using MLP classifier. Overall, combination of both lung and mediastinum features has provided better performance than individual features. Hence, it can be concluded that consideration of mediastinum features along with lungs can enhance the performance of a CAD system to differentiate DS, MDR and XDR TB.

The proposed CAD system involving feature extraction and classification (training and testing) is evaluated with a running time of approximately 4 seconds for all the images in the considered dataset. The results are generated using a system with Intel i5 processor, 8 GB RAM and MATLAB® 2021a software. This approach could be adopted in clinical decision making for screening of DS and DR TB patients.

In literature, limited studies have been reported for classification of drug resistant TB conditions by considering different regions, features and machine learning techniques. Performance of the proposed CAD system has been compared with other existing studies that carried out the differentiation of drug sensitive and resistant TB forms. The methods are compared in-terms of the considered classes, imaging modality, region, features, accuracy, F-measure and AUC as shown in Table 3.

Cheng et al. [12] have investigated the CT features for detection of drug resistant TB. Logistic regression analysis is performed on the significant clinical features and obtained AUC and accuracy of 82.6% and 80.6%, respectively, for differentiation of DS and DR TB. Ureta and Shrestha [32] utilized Convolutional Neural Network (CNN) to classify these conditions using the TB Portal dataset and achieved AUC of 67%. Tulo et al. [25] proposed an automated method for differentiation of DR TB using only mediastinal features from CXRs. The study achieved performance of greater than 93% for classification of DS vs DR TB. However, these studies do not provide any information regarding the classification of MDR and XDR TB, and have not explored the shape features of lungs. Jaeger et al. [2] proposed a CAD method for discrimination of DS and MDR TB conditions using CXR images. The method analysed shape, texture and edge features of the segmented lungs. It achieved AUC of 66% using shape and texture features, 61% using edge orientation features, and recorded even lower performance while using convolutional neural network and deep learning.

Cid et al. [33] reported that in ImageCLEFtuberculosis 2018 campaign different techniques ranging from deep learning to graph-based approaches have been used for differentiation of DS and MDR TB using CT images. The campaign achieved maximum AUC and accuracy of 61.8% and 61.4%, respectively using features extracted from lungs. It is also reported that ImageCLEFtuberculosis 2017 also aimed at differentiation of these conditions [34]. The study obtained AUC and accuracy of 58.3% and 56.8%, respectively. It has been mentioned that, the methods suffer from difficulties in differentiation of drug resistant TB. Kovalev et al. [35] have used textural and shape features of segmented lungs from both CXR and CT images to differentiate DS from DR TB condition. The study is able to obtain limited accuracy of 72.0% by using the combined

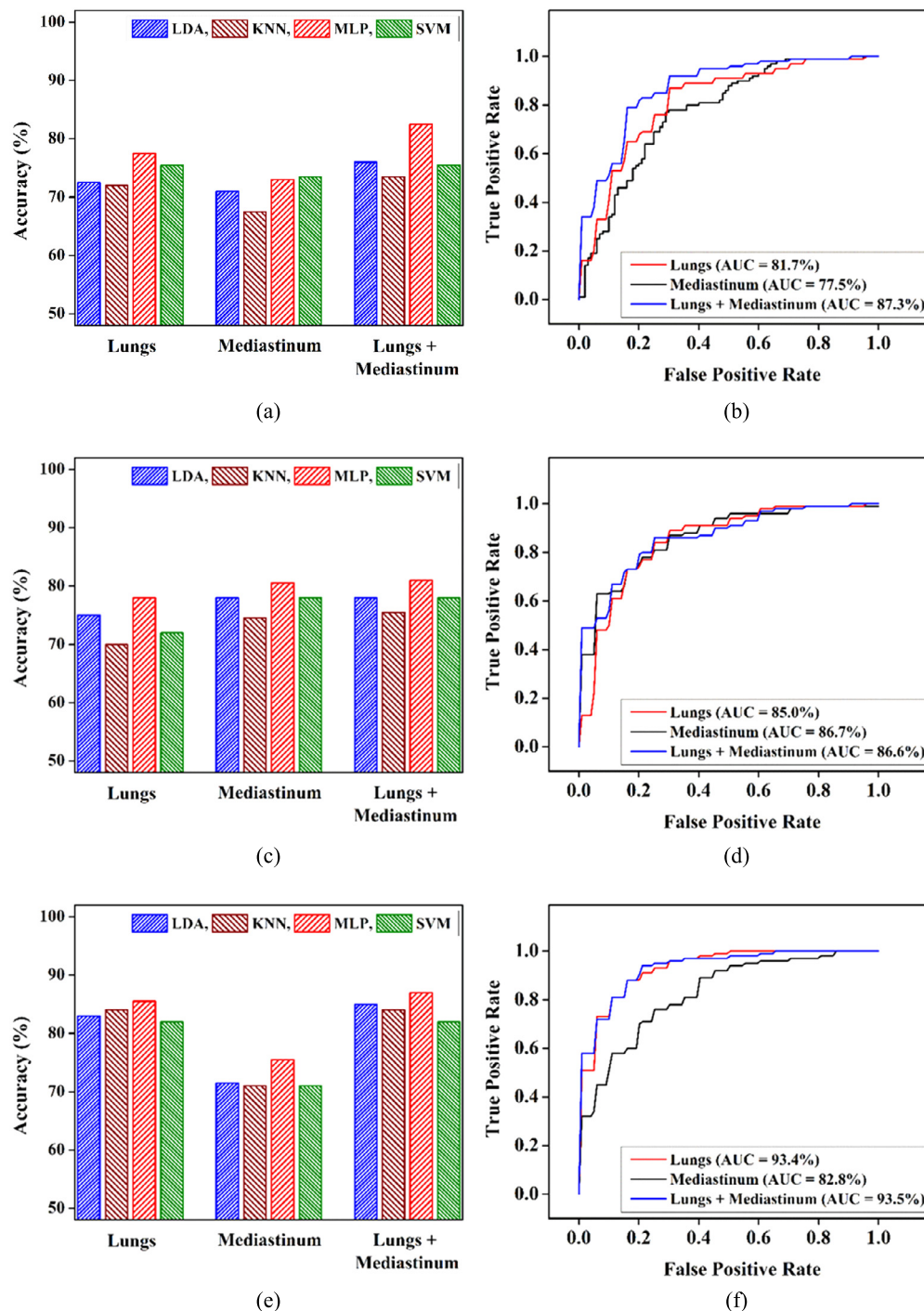


Fig. 7. (a, c, e) Accuracy and (b, d, f) ROC analysis of classifiers for differentiation of DS vs MDR, MDR vs XDR and DS vs XDR TB, respectively.

features from both CXR and CT images. However, the study does not provide any insight on differentiation of drug resistant conditions.

In this work, three binary classifications of TB conditions namely, DS vs MDR, MDR vs XDR and DS vs XDR TB have been performed using lung and mediastinum shape features from CXR images. Here, individual and combination of both lung and mediastinum features are analysed to improve the diagnostic performance of the system. From the results, it is found that consideration of mediastinum shape features along with lung features has improved the performance compared to individual features. The study has achieved AUC of 87.3%, 86.6% and 93.5% for differentiation of DS vs MDR, MDR vs XDR and DS vs XDR TB, respectively.

The proposed system offers a balanced performance for classification of TB conditions in-terms of accuracy, sensitivity, precision, F-measure and AUC. Also, the pipeline of using both mediastinum and lungs shape features has been attempted for the first time to differentiate DR TB conditions. Moreover, the shape feature extraction method is computationally less intense and offers significant feature sets. The proposed work achieves high classification performance as compared to other reported methods.

4. Conclusions

Computer based differentiation of drug sensitive, multi drug resistant and extensively drug resistant TB using chest X-rays helps

Table 2
Maximum Performance of classifiers (in %) for differentiation of TB conditions.

Classes	Features	Classifier	Accuracy	Sensitivity	Precision	F-measure
DS vs MDR	Lungs	MLP	77.5	77.5	77.6	77.5
	Mediastinum	SVM	73.5	73.5	73.7	73.4
	Lungs + Mediastinum	MLP	82.5	82.5	83.1	82.4
MDR vs XDR	Lungs	MLP	78.0	78.0	78.1	78.0
	Mediastinum	MLP	80.5	80.5	80.5	80.5
	Lungs + Mediastinum	MLP	81.0	81.0	81.1	81.0
DS vs XDR	Lungs	MLP	85.5	85.5	85.8	85.5
	Mediastinum	MLP	75.5	75.5	75.8	75.4
	Lungs + Mediastinum	MLP	87.0	87.0	87.2	87.0

Bold indicate best values obtained.

Table 3
Performance comparison of the proposed methodology with existing studies.

Studies	Classes	Modality	Regions	Features	Accuracy (%)	F-measure (%)	AUC (%)
Cheng et al. [12]	DS vs DR	CT	Whole	Clinical	80.6	-	82.6
Ureta and Shrestha [32]	DS vs DR	CXR	Whole	CNN	-	-	67.0
Tulo et al. [25]	DS vs DR	CXR	Mediastinum + Lungs	Shape	-	93.6	-
Jaeger et al. [2]	DS vs MDR	CXR	Lungs	Texture, Shape and Edge	62.0	61.0	66.0
Cid et al. [33]	DS vs MDR	CT	Lungs	Texture and Shape	61.4	-	61.8
Cid et al. [34]	DS vs MDR	CT	Lungs	Texture	56.8	-	58.3
Kovalev et al. [35]	DS vs DR	CXR	Lungs	Texture and Shape	61.7	-	-
Kovalev et al. [35]	DS vs DR	CXR + CT	Lungs	Texture and Shape	72.0	-	72.3
Proposed work	DS vs MDR	CXR	Mediastinum + Lungs	Shape	82.5	82.4	87.3
	MDR vs XDR	CXR	Mediastinum + Lungs	Shape	81.0	81.0	86.6
	DS vs XDR	CXR	Mediastinum + Lungs	Shape	87.0	87.0	93.5

Bold indicate the best values in the category.

in early identification of resistant status of the patient and mass screening of the disease. Analysis of lungs and mediastinum could provide the information about the occurrence of DR TB infection. This work aims to analyze the shape characteristics of both lungs and mediastinum to improve the diagnostics accuracy for differentiation of DS, MDR and XDR TB conditions using CXR based CAD system. In order to do this, the lung and mediastinum shape features are extracted from the segmented masks and machine learning techniques are employed to differentiate the TB conditions.

The extracted shape features are found to be statistically significant for differentiation of the TB conditions. The lung features provide better classification performance for differentiation of DS vs MDR and DS vs XDR, while the mediastinum features are found to be more effective for differentiation of MDR vs XDR TB. As shown by the performance of all three binary classifications of TB conditions, the combination of mediastinum features with the lung features has improved the diagnostic performance. Hence, the proposed methodology involving shape analysis of both lungs and mediastinum appears as a useful computer aided diagnosis system for differentiation of drug sensitive and resistant Tuberculosis conditions. The possibility of characterizing lungs and mediastinum

using other features that represent the structural information of images could be explored in future studies.

Disclosure of interest

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Author contributions

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CRediT authorship contribution statement

S.K. Tulo: Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **P.**

Ramu: Supervision, Validation, Writing – original draft. **R. Swaminathan:** Conceptualization, Investigation, Supervision, Validation, Writing – original draft.

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